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The downloaded binary packages are in
/var/folders/tn/tj4_th0577vb43zq_58mwbm0000gn/T//RtmpEoLuP7/downloaded_packages
> library(readxl)
> data <- read_excel('/Users/chrislee/Desktop/R Workspace/ENBC 332 - Homework2/ENBC 332 - HW02 Problem1 Excel File.xlsx', sheet = "Sheet1")

> # Part A
> covar_for <- function(temp_data) {
+   total_dev <- 0
+   mean_age <- mean(temp_data$age)
+   mean_charges <- mean(temp_data$charges)
+   col_length <- length(temp_data$age)
+   for (i in 1:col_length) {
+     total_dev <- total_dev + (temp_data$age[i] - mean_age) * (temp_data$charges[i] - mean_charges)
+   }
+   print(total_dev)
+   covariance <- total_dev / (col_length - 1)
+   return(covariance)
+ }
> covar_sum <- function(temp_data) {
+   total_dev <- 0
+   mean_age <- mean(temp_data$age)
+   mean_charges <- mean(temp_data$charges)
+   col_length <- length(temp_data$age)
+   total_dev <- sum((temp_data$age - mean_age) * (temp_data$charges - mean_charges))
+   print(total_dev)
+   covariance <- total_dev / (col_length - 1)
+   return(covariance)
+ }
> cat("Covariance for dataset using for loop:", covar_for(data))
[1] 68019611
Covariance for dataset using for loop: 50874.8
> cat("Covariance for dataset using sum function:", covar_sum(data))
[1] 68019611
Covariance for dataset using sum function: 50874.8

> # Part B
> ### Previous functions from homework 2
> var_func_for <- function(data_col) {
+   data_mean <- mean(data_col)
+   total_subtracted_mean <- 0
+   for (data_point in data_col) {
+     total_subtracted_mean <- total_subtracted_mean + (data_point - data_mean)^2
+   }
+   data_variance <- total_subtracted_mean / (length(data_col) - 1)
+   return(data_variance)
+ }
> var_func_sum <- function(data_col) {
+   data_mean <- mean(data_col)
+   subtracted_mean_values <- (data_col - data_mean)
+   squared_difference <- subtracted_mean_values^2
+   total_subtracted_mean <- sum(squared_difference)
+   data_variance <- total_subtracted_mean / (length(data_col) - 1)
+   return(data_variance)
+ }
> sd_func_for <- function(data_col) {
+   data_mean <- mean(data_col)
+   total_subtracted_mean <- 0
+   for (data_point in data_col) {
+     total_subtracted_mean <- total_subtracted_mean + (data_point - data_mean)^2
+   }
+   data_standard_deviation <- sqrt(total_subtracted_mean / (length(data_col) - 1))
+   return(data_standard_deviation)
+ }
> sd_func_sum <- function(data_col) {
+   data_mean <- mean(data_col)
+   subtracted_mean_values <- (data_col - data_mean)
+   squared_difference <- subtracted_mean_values^2
+   total_subtracted_mean <- sum(squared_difference)
+   data_standard_deviation <- sqrt(total_subtracted_mean / (length(data_col) - 1))
+   return(data_standard_deviation)
+ }
> cor_coe_for <- function(temp_data) {
+   cor_coe <- covar_for(temp_data) / (sd_func_for(temp_data$age) * sd_func_for(temp_data$charges))
+   return(cor_coe)
+ }
> cor_coe_sum <- function(temp_data) {
+   cor_coe <- covar_sum(temp_data) / (sd_func_sum(temp_data$age) * sd_func_sum(temp_data$charges))
+   return(cor_coe)
+ }
> cat("Pearson's correlation coefficient for the dataset using for loops:", cor_coe_for(data))
[1] 68019611
Pearson's correlation coefficient for the dataset using for loops: 0.2990082
> cat("Pearson's correlation coefficient for the dataset using sum function:", cor_coe_sum(data))
[1] 68019611
Pearson's correlation coefficient for the dataset using sum function: 0.2990082

> # Part C
> plot(data$age, data$charges,
+       main="Scatter Plot of Coffee Charges in Relation to Age",
+       xlab="Age",
+       ylab="Charges",
+       pch=19)
> |

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